Лабораторная работа №4

Операционные системы

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Информация

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1. Цель работы

Освоение продвинутых методов работы с git-репозиториями и механизмами создания релизов.

2. Задание

- Выполнить работу для тестового репозитория.
- Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

3. Теоретическое введение

Gitflow Workflow, опубликованная Винсентом Дриссеном, предполагает строгую модель ветвления с учетом выпуска проекта и включает создание отдельной ветки для исправления ошибок в рабочей среде. Семантическое версионирование (SemVer) задается в формате МАЖОРНАЯ.МИНОРНАЯ.ПАТЧ, где мажорная версия увеличивается при несовместимых изменениях API, минорная — при добавлении новой обратно совместимой функциональности, а патч-версия — при обратно совместимых исправлениях. Conventional Commits — это соглашение о структуре сообщений коммитов, которое тесно связано с SemVer и регламентирует основные типы коммитов.

4. Выполнение лабораторной работы

Устанавливаю nodejs, пакетный менеджер для него pnpm и gitflow. (рис. 1)

```
 nodejs-docs                                          noarch        1:22.22.0-2.fc43
 nodejs-full-i18n                                     x86_64        1:22.22.0-2.fc43
 nodejs-npm                                           x86_64        1:10.9.4-1.22.22.0.2.fc43

Transaction Summary:
Installing:         6 packages

Total size of inbound packages is 42 MiB. Need to download 42 MiB.
After this operation, 230 MiB extra will be used (install 230 MiB, remove 0 B).
Is this ok [y/N]: y
(1/6] nodejs-1:22.22.0-2.fc43.x86_64
(2/6] pnpm-0:10.27.0-1.fc43.noarch
(3/6] nodejs-npm-1:10.9.4-1.22.22.0.2.fc43.x86_64
(4/6] nodejs-docs-1:22.22.0-2.fc43.noarch
(5/6] nodejs-full-i18n-1:22.22.0-2.fc43.x86_64
(6/6] nodejs-libs-1:22.22.0-2.fc43.x86_64
--------------------------------------------------------------------------------
(6/6] Total
Running transaction
(1/8] Verify package files
(2/8] Prepare transaction
(3/8] Installing nodejs-libs-1:22.22.0-2.fc43.x86_64
(4/8] Installing nodejs-1:22.22.0-2.fc43.x86_64
(5/8] Installing pnpm-0:10.27.0-1.fc43.noarch
(6/8] Installing nodejs-npm-1:10.9.4-1.22.22.0.2.fc43.x86_64
(7/8] Installing nodejs-full-i18n-1:22.22.0-2.fc43.x86_64
(8/8] Installing nodejs-docs-1:22.22.0-2.fc43.noarch
Complete!
mamartynova@mamartynova:~$ wget -q https://raw.githubusercontent.com/petervanderdoes/gitflow-avh/develop/contrib/gitflo
```

Рис. 1: Устновка необходимого ПО

Устаналиваю через pnpm commitizen и standard-changelog.(рис. 2)

```
root
WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +149
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Progress: resolved 149, reused 0, downloaded 149, added 149, done

/home/mamartynova/.local/share/pnpm/global/5:
+ commitizen 4.3.1

Done in 21.6s using pnpm v10.27.0
mamartynova@mamartynova:~$ pnpm add -g standarg-changelog
ERR_PNPM_FETCH_404  GET https://registry.npmjs.org/standarg-changelog: Not Found - 404

This error happened while installing a direct dependency of /home/mamartynova/.local/share/pnpm/global/5

standarg-changelog is not in the npm registry, or you have no permission to fetch it.

No authorization header was set for the request.
Progress: resolved 1, reused 1, downloaded 0, added 0
mamartynova@mamartynova:~$ pnpm add -g standard-changelog
WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +32
++++++++++++++++++++++++++++++
Progress: resolved 181, reused 149, downloaded 32, added 32, done

/home/mamartynova/.local/share/pnpm/global/5:
+ standard-changelog 7.0.1

Done in 11.8s using pnpm v10.27.0
mamartynova@mamartynova:~$
```

Рис. 2: Уствновка через pnpm

Создаю новый репозиторий и делаю там первый коммит. (рис. 3)

```
amartynova@mamartynova:~$ gh repo create --public git-extended
 Created repository Lanawx/git-extended on github.com
 https://github.com/Lanawx/git-extended
amartynova@mamartynova:~$ git clone --recursive git@github.com:Lanawx/git-extended.git
loning into 'git-extended'...
arning: You appear to have cloned an empty repository.
amartynova@mamartynova:~$ ls
esktop  Documents  Downloads  git-extended  gitflow  Music  Pictures  Public  Templates  Videos  work
amartynova@mamartynova:~$ cd git-extended/
amartynova@mamartynova:~/git-extended$ touch README.md && echo text > README.md
amartynova@mamartynova:~/git-extended$ git add /
atal: /: '/' is outside repository at '/home/mamartynova/git-extended'
amartynova@mamartynova:~/git-extended$ git add .
amartynova@mamartynova:~/git-extended$ git commit -m "first commit"
main (root-commit) dce2551] first commit
1 file changed, 1 insertion(+)
create mode 100644 README.md
amartynova@mamartynova:~/git-extended$ git push
numerating objects: 3, done.
ounting objects: 100% (3/3), done.
riting objects: 100% (3/3), 874 bytes | 874.00 KiB/s, done.
otal 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
o github.com:Lanawx/git-extended.git
 * [new branch]      main -> main
amartynova@mamartynova:~/git-extended$
```

Рис. 3: Создание нового репозитория

Инициализирую и конфигурирую общепринятые коммиты в созданной директории через редактирование package.json.(рис. 4)

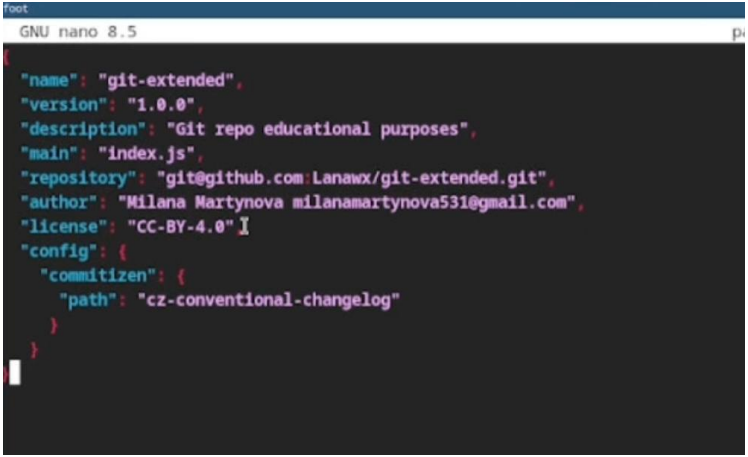

Рис. 4: Конфигурация package.json

Делаю снимок измененний, создаю коммит и отправляю на удаленный репозиторий. (рис. 5)

```
mamartynova@mamartynova:~/git-extended$ nano package.json
mamartynova@mamartynova:~/git-extended$ git add .
mamartynova@mamartynova:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: docs:     Documentation only changes
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 94 chars):
 (27) improve information content
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[main 9edb162] docs: improve information content
 1 file changed, 16 insertions(+)
 create mode 100644 package.json
mamartynova@mamartynova:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.16 KiB | 236.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Lanawx/git-extended.git
   dce2551..9edb162  main -> main
```

Рис. 5: Отправление коммита на github

Инициализирую в репозитории git flow и создаю 1 релиз в только что созданной ветке develop. (рис. 6)

```
foot
mamartynova@mamartynova:~/git-extended$ git flow init

Which branch should be used for bringing forth production releases?
   - main
Branch name for production releases: [main]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/mamartynova/git-extended/.git/hooks]
mamartynova@mamartynova:~/git-extended$ git branch
* develop
  main
mamartynova@mamartynova:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:      https://github.com/Lanawx/git-extended/pull/new/develop
remote:
To github.com:Lanawx/git-extended.git
 * [new branch]      develop -> develop
mamartynova@mamartynova:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
mamartynova@mamartynova:~/git-extended$ git flow release start 1.0.0
```

Создаю список изменений через standard changelog, заканчиваю релиз и выгружаю на удаленный репозиторий изменения. (рис. 7)

```
1 file changed, 4 insertions(+)
 create mode 100644 CHANGELOG.md
mamartynova@mamartynova:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 CHANGELOG.md
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 4 ++++
 1 file changed, 4 insertions(+)
 create mode 100644 CHANGELOG.md
Deleted branch release/1.0.0 (was 4afc159).

Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'main'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'
```

Инициализирую ветку feature для работы над новой функциональностью, готовлю релиз и загружаю на github. (рис. 8)

```
foot
- When done, run:

    git flow release finish '1.2.3'

mamartynova@mamartynova:~/git-extended$ nano packagee.json
mamartynova@mamartynova:~/git-extended$ nano package.json
mamartynova@mamartynova:~/git-extended$ standard-changelog
mamartynova@mamartynova:~/git-extended$ git add .
mamartynova@mamartynova:~/git-extended$ git status
On branch release/1.2.3
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   CHANGELOG.md
        modified:   package.json

mamartynova@mamartynova:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip) site
? Write a short, imperative tense description of the change (max 87 chars):
 (15) update chngelog
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[release/1.2.3 e896dad] chore(site): update chngelog
 2 files changed, 5 insertions(+), 1 deletion(-)
mamartynova@mamartynova:~/git-extended$ git flow ralease finish 1.2.3
```

Рис. 8: Итоговая выгрузка

5. Выводы

В результате выполнения лабораторной работы были освоены навыки корректной работы с git-репозиториями.